MILITARY SPECIFICATION

MACHINE GUN, 7.62MM: M60E2 (FIXED)

This specification is approved for use by Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 7.62mm, gas-operated, air-cooled, link-belt fed, lightweight machine gun designed for coaxial use with the M60A1 tank. It is fired remotely by means of a solenoid.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-13855	- Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-45662	- Calibration System Requirements

STANDARDS

Federal

Fed. Std. No. 356	- Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105	- Sampling Procedures and Tables for-Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

DRAWINGS

Maremont Corporation
MD 3033 - Machine Gun, 7.62MM, Fixed: M60E2

(Product drawings referenced in this specification form a part of Drawing MD 3033.)

PUBLICATIONS

Rock IslandArsenal
SAPD-263 - Springfield Armory Purchase Description - Solenoid for Machine Gun, 7.62MM; M60C (Fixed)

(CoPies of specifications, standards, drawings and publi cations required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1).Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The machine gun shall conform to the material, construction and design requirements specified herein, on Drawing MD 3033, and in MIL-k&l 3855.

3.2.1 Barrel assembly. The barrel assembly and gas evacuation assembly shall be capable of being manually assembled to and disassembled from the receiver when the barrel lock lever is in the vertical position and the bolt fully retracted. The barrel and gas evacuation assemblies shall be fabricated so that when the machine gun is completely assembled, the targeting and accuracy requirements shall be met. The barrel assembly shall be free of cracks and seams, and the bore, chamber and locking surfaces shall be free of foreign matter, corrosion, burrs, pits, bulges and deformation. No mechanical methods of removing chromium plating are permitted in the tube bore. Burrs and sharp edges shall be removed from chamber edges. Scratches or marks, occurring in a chamber which otherwise meets the surface roughness requirements, shall be permitted provided that they do not cause marks on the case of a high-pressure test cartridge fired in the chamber. The barrel assembly shall be proof and magnetic particle inspection marked in accordance with the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

3.2.1.1 Gas cylinder. The gas cylinder shall be fastened securely on the tube assembly by the gas cylinder rivets without relative movement. With the barrel assembly disassembled from the receiver assembly, the gas piston shall move of its own weight through its full range of travel in the gas cylinder.

3.2.1.2 Gas cylinder extension, obturator and socket cap screw. The gas cylinder extension and nut shall be assembled securely to the gas cylinder without relative movement and shall be retained by the lock wire. The gas cylinder socket cap screw (MS 24678-19), extension and obturator shaH be lock wired as specified on the applicable drawing.

3.2.2 Barrel lock lever. The barrel lock lever shall,move manually between the open and closed positions and shall be retained in either position by spring action until manually changed, and shall retain the barrel assembly in the receiver assembly when set in the closed position.

3.2.3 Bolt assembly. With the cover assembly opened and the bolt assembly disengaged from the operating rod yoke, the bolt assembly shall move of its own weight through its full range of travel in the slideways of the receiver. Firing pin protrusion shall be as specified on the applicable drawing. The bolt assembly shall be proof and magnetic particle inspection marked in accordance with the applicable drawing.

3.2.3.1 Cam actuator assembly. The cam actuator assembly shall-be retained on the bolt by the bolt plug and shall rotate without binding on the bolt. The bolt guide roller and the cam actuator roller shall be retained on the cam actuator by the roller rivet and shall rotate without binding on the cam actuator.

3.2.3.2 Ejector. The ejector shall be retained in the bolt by the staked ejector retainer and shall be held in the forward position by spring action. The ejector shall move through its full range of travel under spring action without binding and shall eject cartridge cases completely out of the machine gun when the machine gun is fired or manually operated.

3.3.3.3 Extractor. The extractor shall be retained in the bolt by the extractor plunger and spring. The extractor shall move through its full range of travel under' spring action without binding and shall extract cartridge cases from the barrel when the machine gun is fired or manually operated.

3.2.3.4 Firing pin. The firing pin shall move through its full range of travel in the bolt under spring action without binding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

3.2.4 Buffer assembly. The buffer assembly shall be retained in The receiver assembly by the buffer retaining yoke. The buffer retaining yoke and buffer assembly shall disassemble from the receiver assembly without the use of tools. The plunger shall move through its full range of travel in the buffer body and shall return to its original position by spring action after partial or full travel.

3.25 Backplate assembly. The backplate assembly shall be retaingd on the receiver by engagement of the buffer assembly with the backplate catch and shall disassemble from the receiver assembly and buffer assembly without the use of tools.

3.2.6 Cartridge feed tray assembly. The cartridge feed tray assembly shall be retained on the receiver by the cover hinge pin.

3.2.6.1 Cartridge retainer pawl. The cartridge retainer pawl shall be retained in the frame assembly by the shaft. The cartridge retainer pawl shall be held normally in position by the spring and shall be returned to the original position by spring action after partial or full travel. The pawl shall move through its full range of travel under spring action without binding and shall retain the linked cartridge in the feed mechanism.

3.2.6.2 Cartridge rollers. The cartridge rollers shall be retained in the frame assembly by the shaft and shall rotate without binding on the shaft.

3.2.7 Charger assembly. The charger assembly shall be retained on the receiver by the charger assembly retainer and bracket. The charger assembly retainer and bracket shall be fastened securely to the receiver without relative movement.

3.2.7.1 Cable assembly. The cable assembly shall move through its full range of travel without binding. The cable assembly shall be held in the forward position by the spring. When retracted, the charger shall engage the operating rod assembly and shall retract the operating rod assembly and bolt assembly. The swaged ends of the cable assembly shall withstand a static load of 150 pounds.

3.2.8 Cover assembly. The cover assembly shall be retained on the receiver assembly by the cover hinge pin. The cover assembl"y shall be capable of being locked in the closed position by the cover latch when the bolt assembly is fully retracted. The cover shall open to the stop position when the cover latch is released, and shall open to full open position when the stop latch is released. The cover shall be retained ‘in both open positions by spring action.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

3.2.8.1 Latch lever assembly. The latch lever assembly shall rotate manually to disengage the cover assembly from the receiver assembly and shall be returned to the stop position by spring action after partial or full travel. The latch lever assembly shall rotate the cover latch to lock and unlock the cover assembly.

3.2.8.2 Cartridge guides. The cartridge guides shall be retained in the cover housing assembly by the cartridge guide shaft and cotter pin. The cartridge guides shall operate through their full range of travel under spring action without binding and shall be positioned to properly guide cartridges into the chamber.

3.2.8.3 Feed cam assembly. The feed cam assembly shall be retained in the cover housing assembly by the feed cam retainer and shall move its full range of travel under spring action without binding. The feed cam assembly shall return to its stop position by spring action after partial or full travel. Movement of the feed cam assembly shall operate the feed pawl assembly.

3.2.8.4 Feed pawl assembly. The feed pawl assembly shall be retained in the cover housing assembly by the chassis rollers and when disengaged from the feed lever assembly shall move of its own weight through its full range of travel. The feed pawl assembly shall be capable of feeding linked cartridges into the feed mechanism.

3.2.9 Operating rod assembly. With the cover assembly opened, the drive spring and guide assembly disassembled from the operating rod assembly, and the sear disengaged; the operating rod assembly, together with the bold assembly, shall slide from the open to closed position by its own weight. The operating rod assembly shall manually assemble to and disassemble from the slideway of the receiver. The yoke and head shall be retained on the tube by staked rivets without relative movement. The roller shall be retained on the yoke by the yoke roller pin and shall rotate without binding. The pin shall be retained securely by the cross pin as specified on the applicable drawing. With the solenoid deenergized and the operating rod retracted into the cocked position, the sear and the operating rod shall be held in the cocked position until the solenoid is energized.

3.2.10 Receiver assembly group The slideways of the receiver assembly shall be smooth and free of burrs.

3.2.11 Actuator assembly, The actuator assembly shall be retained on the receiver assembly by the retaining pin and shall disassemble from the receiver assembly without the use of tools. The retaining pin shall be locked in place by the retaining pin spring lock.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

3.2.11.1 Solenoid. The solenoid shall be as specified in SAPD-263. Unless otherwise specified, the solenoid shall be assembled to the solenoid housing in the position depicted on the applicable drawing. The solenoid shall be securely retained to the solenoid housing by the rim clenching clamp which shall be held in position by the clamp bolt.

3.2.11.2 Sear. The sear shall be capable of full engagement with the sear notch on the operating rod and of holding the operating rod in a cocked position. When the solenoid (see 3.2.11.1), is energized, the sear shall disengage from the sear notch on the operating rod allowing the operating rod to move forward under spring action. When the solenoid is deenergized, the sear shall return to the engaging position by spring action of the sear plunger.

3.2.11.3 Trigger. The trigger shall operate with a smooth, continuous movement when actuated. The trigger push necessary to operate the firing mechanism shall be not less than 2.5 pounds and not more than 11.5 pounds. The trigger shall return to the normal position under spring action of the sear plunger, after partial or complete trigger push.

3.2.11.4 Safety. The safety shall move manually between the safe and fire position, and shall remain in the set position under spring action until reset manually. The trigger mechanism shall operate smoothly, without binding, through its full range of travel when the red surface is readily visible. When the red surface is not visible, the safety shall block all trigger action.

3.2.12 Driving spring and driving spring guide assembly. Driving spring ends shall be securely brazed. Guide assembly stop shall be securely brazed to the rod.

3.3 Performance characteristics.

3.3.1 Firing pin indent. The firing pin indent shall be not less than 0.030 inch nor more than 0.035 inch. The firing pin indent shall not be off center more than one-half the diameter of the firing pin point. Testing shall be as specified in 4.5.3.2.

3.3.2 Headspace. The headspace shall be not less than 1.6315 inches nor more than 1.6365 inches when measured to the .400 diameter datum on the first shoulder of the chamber. Testing shall be as specified in 4.5.3.3.

3.3.3 High-pressure resistance. The machine gun shall be capable of withstanding the firing of one Government Standard 7.62mm, M60 high--pressure test cartridge. Parts shall be free of cracks, seams and other injurious defects after proof firing. Testing shall be as specified in 4.5.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

3.3.4 Functioning firing. The machine gun shall operate without malfunctions or unserviceable parts, shall have a cyclic rate of fire of 500 to 650 rounds per minute, and shall withstand a belt pull of 10 pounds. Testing shall be as specified in 4.5.3.4.

3.3.5 Targeting and accuracy.

3.3.5.1 Targeting. Nine rounds of a ten-round burst fired from the machine gun at a range of 100 meters shall be within the 21.6 x 21.8 inch rectangle in accordance with Drawing 75C21371. Testing shall be as specified in 4.5.3.4.

3.3.5.2 Accuracy. Nine rounds of a ten-round burst fired from the machine gun at a range of 100 meters shall be within or cut the edge of a 12.6 inch diameter circle. Testing shall be as specified in 4.5.3.4.

3.3.6 Endurance. The machine gun shall be capable of firing 10,000 rounds of M59 or M80 ball cartridges, using M13 links, with not more than the number of malfunctions and unserviceable parts allowed in Table I. The average cyclic rate of fire for the entire test shall be within the limits specified in 3.3.4. Testing shall be as specified in 4.5.3.5.

Table I

Malfunctions and Unserviceable Parts

Code	Malfunction	Number permitted in 10,000 rounds
LP	Loosening of parts (failure of bonding and riveting)	0
LPs	Loosening of parts (failure of staking)	1
FF	Failure to feed	4
FEJ	Failure to eject	3
FEX	Failure to extract	1
OM	Other malfunctions	1

Code	Unserviceable parts	Number permitted in 10,000 rounds	
		First 5,000 rounds	Second 5,000 rounds
PAB	Plug, Bolt assembly	0	1
WL	Lock Washer	1	1
S	Solenoid	0	0
OP	Other parts[2]		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

When malfunctions are traceable to particular parts, it is permissible to replace such parts and record them as unserviceable, subject to limitations of Table I. When it is definitely established by the Government representative that previously recorded malfunctions are attributable to an unserviceable part, such malfunctions shall not be counted against the machine gun being tested, provided that they occurred not more than 200 rounds prior to replacement of the unserviceable part. These 200 rounds shall have been fired with the unserviceable part. However, such malfunctions shall remain recorded and properly identified. An unserviceable part is one that causes malfunctions or impairs the safety of the weapon. Malfunctions attributable to links and ammunition shall not be counted against the machine gun. However, such malfunctions shall be recorded.

[2]One unserviceable part other than those specified shall be allowed in the test and shall be subject to the judgement of the Government representative that the failure does not represent an unsafe condition or a defective condition which is prevalent throughout the lot of items involved.

3.4 Interchangeability Unless otherwise specified on the drawings, all parts shall be interchangeable. Testing shall be as specified in 4.5.3.6.

3.5 Marking. Marking shall be in accordance with MIL-W-13855. When lv160C machine guns are modified to M60E2 configuration, the name of the modifying agency shall be placed on the receiver below the name of the original manufacturer.

3.6 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, quality assurance provisions, and the contract. These minimum inspections shall not be construed as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the contract, drawings and specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with contract requirements (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. The number of machine guns in an inspection lot shall be either 200 or one month's production, whichever is smaller. Machine guns shall be assembled from lots of component parts that have met all inspection requirements specified herein. Endurance test lot size shall be as specfied in 4.5.3.5.1.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of Parts shall be Performed in accordance with the criteria specified in the-contract (see 6.1). Examination of parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.5.2.2 Machine guns. Visually and manually examine each machine gun in accordance with the following. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating, to determine the general quality and completeness of manufacture and assembly, marking (see 3.5) and workmanship (see 3.6). The examination provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operation. Reinspection of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. Machine guns failing to meet the requirements shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.5.2.2.1 Barrel assembly.

a. Manually examine functioning of barrel lock lever.
b. Disassemble barrel and gas evacuation assembly. Assure that with barrel lock lever in the vertical position, disassembly from the receiver is readily accomplished (see 3.2.1).
c. Visually examine surface finish of barrel assembly, particularly the bore, chamber, lock surfaces and chromium plating to determine compliance with 3.2.1.
d. Visually examine barrel assembly for presence of proof and magnetic particle inspection marks (see Drawings F7269042 and MD0346).

4.5.2.2.1.1 Gas cylinder.

a. Visually and manually examine gas cylinder to determine compliance with 3.2.1 and 3.2.1.1.
b. Disassemble rear gas cylinder obturator and piston. Visually examine piston for cracks, burrs and deformations. Reassemble assuring that head of piston is assembled toward breech (rear) end.
c. Visually and manually examine gas cylinder cap screw, extension and obturator for secure lock wiring and secure assembly.

4.5.2.2.2 Barrel lock lever. Visually and manually examine the barrel lock lever to determine compliance with 3.2.2.

4.5.2.2.3 Bolt assembly.

a. Visually and manually examine bolt for proper functioning and for presence of proof and magnetic particle inspection marks (see Drawing F11010358).
b. Examine firing pin protrusion using-the inspection equipment in accordance with Drawing MD3033.
c. Visually examine to assure that ejector face is flush with or below front of bolt (see Drawing MD3032) and ejector retaining pin is securely staked.

4.5.2.2.3.1 Cam actuator assembly. Manually examine cam actuator assembly to determine compliance with 3.2.3.1.

4.5.2.2.3.2 Ejector. Manually examine retention and spring action of ejector to determine compliance with 3.2.3.2.

4.5.2.2.3.3 Extractor. Manually examine retention and spring action of extractor to determine compliance with 3.2.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.5.2.2.3.4 Firing pin. Manually examine firing pin to determine compliance with 3.2.3.4.

4.5.2.2.3.5 Disassembly of bolt assembly.

a. Disassemble all parts except ejector from bolt assembly.
b. Visually examine all parts for mutilations, breaks or cracks.
c. Visually examine bolt lugs, operating cam cut, rollers, guideways, cam actuator assembly, extractor lips and ejector face for cracks, burrs, sharp edges or deformations.
d. Visually examine firing pin striker point; it shall be smooth and free of pits and burrs. Examine fillet radii at spools for cracks and deformations.
e. Reassemble bolt assembly, assuring that cam actuator assembly is properly assembled with roller end forward on bolt.

4.5.2.2.4 Buffer and back plate assemblies. Visually and manually examine buffer and back plate assemblies to determine compliance with 3.2.4 and 3.2.5 respectively.

4.5.2.2.5 Cartridge feed tray assembly.

a. Visually examine cartridge feed tray assembly to determine compliance with 3.2.6.
b. Visually examine feed plate guideways and cartridge stops for burrs, sharp edges or deformations.

4.5.2.2.5.1 Cartridge retainer pawl.

a. Visually and manually examine retention, spring action, and free travel of cartridge retainer pawl to determine compliance with 3.2.6.1.
b. Visually examine the cartridge retaining pawl for cracks, burrs, sharp edges or deformations.

4.5.2.2.5.2 Cartridge rollers. Visually and manually examine cartridge rollers to determine compliance with 3.2.6.2. .

4.5.2.2.6 Charger assembly. Manually examine charger to determine compliance with 3.2.7.

4.5.2.2.6.1 Cable assembly.

a. Manually examine retention and free movement of cable assembly to determine compliance to 3.2.7.1.
b. Visually and manually examine the swaging to determine compliance with 3.2.7.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.5.2.2.7 Cover assembly. Visually and manually examine cover assembly to determine compliance with 3.2.8.

4.5.2.2.7.1 Latch lever assembly Manually examine spring action of latch lever assembly to determine compliance with 3.2.8.1.

4.5.2.2.7.2 Cartridge guides.

a. Visually and manually examine retention and spring action of cartridge guides to determine compliance with 3.2.8.2.
b. Visually examine cartridge guides for cracks, burrs, sharp edges or deformations.

4.5.2.2.7.3 Feed cam assembly.

a. Visually and manually examine feed cam assembly to determine compliance with 3.2.8.3.
b. Visually examine feed cam assembly for burrs, cracks, sharp edges or deformations.

4.5.2.2.7.4 Feed pawl assembly.

a. Visually and manually examine retention and free travel of feed pawl assembly to determine compliance with 3.2.8.4.
b. Visually examine feed pawl assembly for cracks, burrs, sharp edges or deformations.

4.5.2.2.8 Operating-rod assembly.

a. Visually and manually examine action of operating rod assembly and bolt assembly to determine compliance with 3.2.9.
b. Disassemble operating rod assembly from bolt assembly and visually and manually examine operating rod assembly to determine compliance with 3.2.9.
c. Visually examine rod for wear or mutilations on sear engagement notch.
d. Visually examine yoke for mutilations or burrs on bolt caroming surfaces.
e. Visually examine roller for wear or mutilations.

4.5.2.2.9 Receiver assembly group. Visually and manually examine receiver assembly group to determine compliance with 3.2.10.

4.5.2.2.10 Actuator assembly.

a. Visually and manually examine actuator assembly to determine compliance with 3.2.11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

b. Disassemble pin, sear, spring and plunger from acutuator assembly.

4.5.2.2.10.1 Solenoid.

a. Visually and manually examine solenoid to determine compliance with 3.2.11.1.
b. Visually examine end of armature shaft for burrs and deformations.

4.5.2.2.10.2 Sear.

a. Visually examine electrical function of solenoid and sear action to determine compliance with 3.2.11.2.
b. Manually examine for minimum height of sear catch from top of housing (0.265 inch) with solenoid deenergized.
c. Visually examine sear plunger and spring for burrs, cracks and deformations.

4.5.2.2.10.3 Trigger. Visually and manually examine trigger to determine compliance with 3.2.11.3.

4.5.2.2.10.4 Safety. Visually and manually examine safety to determine compliance with 3.2.11.4.

4.5.2.2.11 Driving spring and driving spring guide assembly.

a. Disassemble driving spring and guide assembly from operating rod assembly and buffer plunger.
b. Visually examine ends of spring for secure brazing (see Drawing B7269303) to determine compliance with 3.2.12.
c. Visually examine guide assembly for secure brazing of stop to rod (see Drawing C7269199) to determine compliance with 3.2.12.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete machine gun. If test requirements cited herein are not met, acceptance of the machine gun shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

c. Examine machine guns, partially assembled machine guns, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned subassembly for retest.

4.5.3.2 Firing pin indent testing. Ten machine guns selected at random from each inspection lot shall be tested for firing pin indent (see 3.3.1) using the test method specified in 4.6.1. Failure of any machine gun in the sample to meet the requirements shall cause rejection of the represented lot.

4.5.3.3 Heads ace testing Each machine gun shall be tested for headspace (see 3.3.2 using the test method specified in 4.6.2. Machine guns failing to meet the requirements shall be rejected.

4.5.3.4 High-pressure resistance, functioning firing, and targeting and accuracy testing. Each machine gun shall be tested for high-pressure resistance (see 3.3.3), functioning firing (see 3.3.4) and targeting and accuracy (see 3.3.5), using the test methods specified in 4.6.3, 4.6.4 and 4.6.5, respectively. Machine guns failing to meet any of the requirements shall be rejected. Also, if machine guns are fired for targeting and accuracy at a range of less than 100 meters, a sample of five machine guns (which have passed the test at the shorter range) from each inspection lot shall be tested at a range of 100 meters using the test methods specified in 4.6.5. Failure of any machine gun in the sample to meet the requirements shall cause rejection of the represented lot.

4.5.3.5 Endurance testing.

4.5.3.5.1 Lot size. The first five endurance test lots shall each consist of 200 machine guns or a month’s production, whichever is smaller. When five successive lots meet the endurance requirements, the lot size shall be increased to 500 machine guns or a months production, whichever is smaller. When five successive lots of the increased size have met the endurance requirements, the lot size shall be further increased to 1,000 machine guns or a month's production, whichever is smaller. If rejection of a lot occurs at any time, the next smaller test lot size criteria shall be reinstated and the above procedure repeated in returning to the larger lot size.

4.5.3.5.2 Procedure. One machine gun randomly selected from each endurance lot shall be tested for endurance (see 3.3.6) using the test methods specified in 4.6.6. The contractor shall provide replacement

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

parts as required to complete the test at no additional cost to the Government. If the endurance requirements are not met, the represented lot shall be rejected subject to retest or reconditioning and further test as a reconditioned lot. An endurance retest of two other machine guns from the same lot shall be made without reconditioning the represented lot unless, in the opinion of the Government representative, the failure indicates serious defects in the item, in which case retest shall be made only when authorized by the procuring agency. Failure of either machine gun in the retest to meet the requirements shall cause rejection of the represented lot subject to reconditioning and further testing as a reconditioned lot. Prior to submission of a lot of machine guns as a reconditioned lot, the cause of failure shall be determined and the contractor correction shall be effected on all machine guns in the lot. Sample size and test methods for reconditioned lots shall be the same as for retest.

4.5.3.6 Interchangeability testing.

4.5.3.6.1 In Plant.

4.5.3.6.1.1 Machine guns. Ten machine guns, selected at random from each inspection lot, shall be tested for interchangeability (see 3.4) using the test method specified in 4.6.7.1.1. Machine guns taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. Test frequency may be reduced to not less than one test of ten machine guns each month after three consecutive inspection lots have been accepted. If rejection of a lot occurs, reinstate above procedure. The ten machine guns shall be tested for firing pin indent (see 4.6.1) before interchange of parts. After interchange of parts, the ten machine guns shall be tested for firing pin indent, headspace, functioning firing and targeting and accuracy using the test methods specified in 4.6.1, 4.6.2, 4.6.4 and 4.6.5, respectively. No failures shall be allowed. Failure of the interchangeability test shall cause retest or rejection of the represented lot. At the discretion of the Government, an interchangeability retest may be allowed without reconditioning the lot of machine guns. Failure in the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of 20 machine guns from each retest or reconditioned lot shall be tested using the same procedure described above.

4.5.3.6.1.2 Concurrent repair parts. At least two parts from each inspection *lot* of concurrent repair parts shall be subjected to the interchangeability test specified in 4.6.7.1.2. Failure of any part to meet the requirements shall be cause for rejection of the represented lot of parts subject to reconditioning and further test as a reconditioned lot. A sample of double the number of parts used in the original test shall be tested from each reconditioned lot using the test method specified in 4.6.7.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.5.3.6.2 Interplant. When machine guns are manufactured concurrently by more than one contractor, each contractor shall forward monthly six machine guns, for the interplant interchangeability test specified in 4.6.7.2 (see 6.1). The contractor will be informed of any failure of the machine guns to meet prescribed requirements.

4.5.3.7 Certification. Unless otherwise specified, the contractor shall furnish the Government representative with certified statements that each inspection lot conforms to the materials and processes specified on the applicable drawings and specifications.

4.5.3.8 Component parts and concurrent repair parts testing. Raw material testing, part testing and certification shall be performed in accordance with the criteria specified in the contract (see 6.1). This will include chemical analysis and physical tests of materials, and tests of protective finish, heat treatment, bonding and function of parts, as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, calibration, maintenance and disposition. Unless otherwise specified (see 6.1), responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on the applicable Quality Engineering Data List and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607 and MIL-C-45662. The applicable Quality Engineering Data List shall be as specified in the contract (see 6.1).

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements, by various experienced inspection/test personal, to an accuracy of ten percent of the total tolerance of the characteristic being inspected.

4.5.5 Packaging examination and testing. Unless otherwise specified (see 6.1), packaging examination and testing shall be in accordance with MIL-P-14232.

4.6 Test methods.

4.6.1 Firing pin indent test. Each sample machine gun shall be tested for firing pin indent, with both its assigned and spare barrel assembly, using inspection equipment in accordance with Drawings D7274381 and C7318984. The machine gun shall be cocked and then held in a horizontal position with the cover assembly opened. The holding fixture containing the copper compression cylinder shall be inserted into the barrel chamber, the cover closed, and the solenoid armature manually

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

depressed to release the bolt and indent the copper cylinder. The holding fixture shall be removed from the machine gun and the depth of the indent in the copper cylinder computed by measuring the distance from the original surface of the copper cylinder (before indentation) to the bottom of the firing pin impression. If any firing pin indent is not within the requirements, three more impressions shall be taken and the average depth of the three indents of each test shall be within the requirements. All firing pin indent impressions shall not be off center more than one-half the diameter of the firing pin point.

4.6.2 Headspace test. Machine guns shall be tested for headspace after proof firing in accordance with Drawing B7274136 using a contractor designed Government approved holding fixture in lieu of F7274459 and D7274515.

4.6.3 High-pressure resistance test. Machine guns shall be tested by firing one high-pressure test cartridge. Testing shall be accomplished with the machine gun mounted upright in a contractor designed, Government approved firing fixture and with the safety shield in position over the machine gun. Each barrel assembly shall have been proof fired and magnetic particle inspected in accordance with the applicable drawing before assembly to the machine gun for this test. Bolts and barrel assemblies to be assembled into production machine guns shall be proof fired only as part of the completed machine guns and magnetic particle inspected and marked in accordance with the applicable drawing and other applicable documents. Bolts and barrel assemblies to be used as repair parts shall be proof fired independently of the production machine guns and magnetic particle inspected and marked in accordance with the applicable drawings and other applicable documents. After proof firing, machine guns shall be examined for cracks, deformations and other evidence of damage and cartridge cases shall be visually examined for bulges, splits, rings and other defects caused by defective barrels. Proof marks shall be applied as indicated on the applicable drawings on machine guns that have passed this test.

4.6.4 Functioning firing test.

4.6.4.1 Machine guns shall be tested in accordance with the schedule specified in Table II. Testing shall be accomplished with the machine gun mounted upright in a contractor designed, Government approved firing fixture.

Table II

Functional Firing Test Schedule

First machine guns	All other machine guns	All other firing.
50 rounds.	20 rounds	Interrupted bursts[1]
50 rounds	20 rounds	One continuous burst[2]

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

[1]Firing shall be intentionally stopped by deenergizing the solenoid at least five times during the interrupted burst firing. The machine gun shall be checked to assure that the firing stops immediately when the current is discontinued.

[2]Rate of fire shall be measured and recorded on every tenth machine gun during the one continuous burst firing using a electronic counting indicator, or approved equal. Failure of any sample machine gun to meet the cyclic rate of fire requirement shall be cause for measurement of the cyclic rate of fire on the other nine machine guns represented.

4.6.4.2 For the first 500 machine guns, the ammunition shall be linked together in 100 round belts with a dummy cartridge separating each 50 rounds. For all other machine guns the ammunition shall be linked together in 100 round belts with a dummy cartridge separating each 20 rounds.

4.6.5 Targeting and accuracy firing test. Machine guns shall be boresighted and fired with the machine gun mounted upright in a contractor designed, Government approved firing fixture.

4.6.5.1 The machine gun shall be boresighted in the following manner:

a. Fully retract bolt.
b. Move barrel lock lever to open position.
c. Disassemble barrel assembly from receiver.
d. Assemble boresight telescope gage conforming to Drawing C7797808 to boresight expanding mandrel gage conforming to Drawing D7796858.
e. Insert mandrel with boresight into barrel socket hole in receiver.
f. Move barrel lock lever to closed position.
g. Boresight machine gun on sighting image depicted on Drawing 75C21371.

4.6.5.2 With the mount locked in the foresighted position, the mandrel with boresight removed, and the barrel assembly reassembled to the receiver, a 10-shot burst shall be fired for the targeting and accuracy requirements using the solenoid.

4.6.5.3 Machine guns may be fired at a range of not less than 25 meters with suitable correlations in the targeting and accuracy requirements as indicated on the targeting and accuracy diagram, Drawing 75C21371, and as approved by the contracting officer.

4.6.6 Endurance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.6.6.1 Machine guns shall be tested mounted upright in a contractor designed, Government approved firing fixture. Ammunition shall be fed to the machine gun from the ammuunition box using 100 round belts.

4.6.6.2 A total of 10,000 rounds shall be fired in the initial test machine gun from each test lot regardless of the number of malfunctions or unserviceable parts requiring replacement in order to complete the test.

4.6.6.3 The machine gun shall be fired in series of approximately 200 rounds. The first 100 rounds shall be fired in interrupted bursts and the second 100 rounds shall be fired in one continuous burst. Firing shall be intentionally stopped (by deenergizing the solenoid) at least 10 times during the interrupted firing of 100 rounds. The machine gun shall be checked to assure that the firing stops immediately when the current to the solenoid is discontinued.

4.6.6.4 The machine gun shall be cooled to ambient temperature after each 200 round series using cooling aids other than water. The cyclic rate of fire shall be measured and recorded on each fifth series. It shall be permissible to clean and oil the machine gun at intervals of not less than 2,000 rounds. At the close of each day's firing, the machine gun shall be protected against corrosion. No parts shall be altered and only parts broken or worn to the extent that they are unserviceable shall be replaced.

4.6.6.5 Complete, accurate records shall be kept for each endurance test, showing each malfunction and part replacement including the number of the round at which each occurred and corrective action taken.

4.6.6.6 At the completion of the endurance test, machine guns shall be disposed of as specified in the contract (see 6.1).

4.6.7 Interchangeability test.

4.6.7.1 In plant.

4.6.7.1.1 Machine guns. Machine guns shall be tested by disassembling and then reassembling parts using the parts and prearranged system prescribed below. Interchange of parts shall be accomplished by dividing the parts of each machine gun into 10 groups of nonmating parts as shown below and distributing the groups into 10 different trays until each tray contains a complete machine gun. Groups of parts from machine gun number 1 shall be taken in order and placed in trays 1 through 10; groups of parts from machine gun number 2 shall be taken in order and placed in trays 2 through 10 to 1; groups of parts from machine gun number 3 shall be taken in order and placed in trays 3 through 10 to 2, etc. Commercial parts such as screws, nuts, washers and pins shall be placed in the same tray as their mating or associate part. Any corrrnercial part rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test. The machine guns shall be reassembled using only those parts which are in the same tray.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

Groups of Nonmating Parts

Group I

Actuator assembly cam (7269063)
Barrel subassembly (MD0346)
Bearing, firing pin (7269065)
Cam assembly, feed (11699814)
Clip, spring lever (7269146)
Extension, gas cylinder (MC3049)

Group II

Receiver assembly (75F20930)
Roller, cartridge feed (7269333)
Ejector (11010375)

Group III

Housing, solenoid (MD0951)
Cover, assembly (7269114)
Extractor (7790907)
Guide assembly, driving spring (7269199)
Charger assembly (MC3095)

Group IV

Rod assembly, operating (MD0851)
Shaft, cartridge guide (11010152)
Shaft, cartridge pawl (7790724)
Spring, firing pin (7269087)
Spring, helical, compression (7269086)

Group V

Guide, cartridge front (7269116)
Retainer (MC0973)
Buffer assembly (MD0969)
Latch, hinge pin (7269243)
Lever, barrel lock (1101084)
Lever, feed (7269145)
Spring, helical, compression (7269211)

Group VI

Pin, hinge cover (7269247)
Plunger, extractor (7269083)
Sear (7269209)
Spring, driving (7269303)

Group VII

Guide, cartridge rear (7269117)
Pawl assembly, feed (7269120)
Pawl, cartridge retainer (7269332)
Pin, firing (11010376)
Pin, retaining (7269205)
Flange, rim clenching (7792092)

Group VIII

Plunger, sear (7269207)
Back plate assembly (MD0986)
Spring, helical, torsion (7269335)
Spring lock, retaining pin (772398)
Bolt (11010358)

Group IX

Piston, gas (7791247)
Plug assembly, bolt (MC0938)
Spring, ejector (75B20827)

Group X

Spring, helical, torsion (7269301)
Tray assembly, cartridge feed (7792096)
Washer, lock (7269035)
Yoke, buffer, retaining (11699786)
Solenoid, electrical (75F20924)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

4.6.7.1.2 Concurrent repair parts. Concurrent repair parts shall be tested by disassembling two machine guns, previously tested in 4.5.3.6.1 as necessary, and then reassembling them using the concurrent repair parts. The machine guns shall operate and function properly. This test may be performed independently of the machine gun interchangeability test specified in 4.5.3.6.1.1, and at more frequent intervals using accepted machine guns taken from current production.

4.6.7.2 Interplant. Machine guns which are to be subjected to the interplant interchangeability test shall be given preliminary hand functioning to assure proper operation before parts are disassembled from the gun, Machine guns shall be interchanged in a manner similar to the detailed plan in 4.6.7.1.1. When disassembling, every other gun used shall be one produced by a different manufacturer. The machine guns shall be tested for and shall comply with the requirements for firing pin indent, headspace, functioning and targeting and accuracy before and after interchange of parts using the test methods specified in 4.6.1, 4.6.2, 4.6.4 and 4.6.5, respectively. Parts shall be identified with their manufacturer throughout the test. Before guns are returned to the contractors, the original parts shall be reassembled to their respective guns and the guns given a hand functioning test to assure proper operation.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack shall consist of a complete machine gun packaged in accordance with the requirements for the level of protection specified in the contract (see 6.1), packed level B and forwarded in accordance with 3.1.

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with the requirements for the level of protection specified in the contract (see 691).

5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

5.4 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the requirements for the level of protection specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. List of drawings, specifications and publications pertinent to the machine gun, showing applicable revision dates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

c. Shipping instructions for first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Inspection criteria for component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.8).
e. Shipping instructions for machine guns and parts when an interplant interchangeability test is required " (see 4.5.3.6.2).
f. Responsibility for acquisition, calibration, maintenance and disposition of inspection equipment, if other than specified (see 4.5.4.1).
g. Packaging inspection, if different (see 4.5.5).
h. Disposition of endurance tested machine guns (see 4.6.6.6).
i. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3).
j. Packaging instructions for repair parts (see 5.4).

Custodian:
Army - WC

Preparing Activity:
Army - WC

Project number:
1005-A553

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-4597A(WC); Machine Gun, 7.62MM: M60E2 (Fixed)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Rock Island Arsenal
ATTN: SARRI-LE-S
Rock Island, Illinois 61201

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

INCH-POUND

MIL-M-45974 (AR)
AMENDMENT 2
18 May 1993
SUPERSEDING
AMENDMENT 1
26 October 1984

MILITARY SPECIFICATION

MACHINE GUN, 7.62MM: M60E2 (FIXED)

This amendment forms a part of MIL-M-45974 (AR), dated 14 May 1976, and is approved for use by the US Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 5

3.2.9 Delete Guide Assembly" and substitute “Drive Spring Guide”.

PAGE 6

3.2.12. Delete in its entirety.

PAGE 7

3.3.5.1. Delete "100 meters" and substitute ‘1OO yards”.

Delete “21.6 x 21.8 inch” and substitute “28.8 inch high x 21.6 inch wide”.

Delete 75C21371” and substitute “12003114”.

3.3.5.2: Delete “100 meters” and substitute “100 yards”.

PAGE 8

* 3.5: Add the following at the end of paragraph 3.5:

“The bar code label shall be firmly affixed to the weapon receiver; the bar coding shall be clearly defined and the number on the label shall agree with the serial number stamped on the receiver.”

DISTRIBUTION STATEMENT A. Approved for public release distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T04:32Z
Check the source to verify that this is the currentversion before use.

PAGE 13

4.5.2.2.11 Delete in its entirety.

PAGE 14

4.5.3.4, lines 7 and 9: Delete "100 meters" and substitute "100 yards".

PAGE 20

GROUP III : Delete 'Guide Assembly, Driving Spring (7269199)" and substitute "Guide, Drive Spring (9362512)".

GROUP VI: Delete "Spring, Driving (7269303)" and substitute. "Spring, Drive (9362511)".

PAGE 22

* "6.1 Add the following at the end of paragraph 6.1:

"K. Bar code marking requirements."

"6.2 Add the following paragraph:

"6.2 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Amament Research and Development Center (AR.DC) may also include drawings prepared by, and identified as U.S. Amy Armament, Research and Development Comand (ARRADCOM) , Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARdc."

The margins of this amendment are marked with an asterisk to indicate where changes (additions~ modifications, correctional deletions) from the previous AMENDMENT were made" This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
ARMY-AR

(Project 1005-A777)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-M-45974(AR)
NOTICE 2
1 December 1995

MILITARY SPECIFICATION

MACHINE GUN, 7.62MM: M60E2 (FIXED)

This notice should be filed in front of ML-M-45974 (AR), dated 14 May 1976

ML-M-45974 (AR), dated 14 May 1976, with Amendment 2, dated 18 May 1993 and Validation Notice 1, dated 12 October 1990, is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity
Army - AR

AMSC N/A

FSC 1005

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:32Z
Check the source to verify that this is the current version before use.